

CS103: Database Management Systems — Page 1: Relational Model, Keys, and Relational Algebra

1. Foundations of the Relational Model

The relational model organizes data into Relations (tables) comprised of Tuples (rows) and Attributes (columns). Each attribute is constrained to an atomic domain of values.

Key Terminology:

- Superkey: A set of one or more attributes that uniquely identifies a tuple within a relation.
- Candidate Key: A minimal superkey (no proper subset is itself a superkey).
- Primary Key: The specific candidate key chosen by the database designer as the principal tuple identifier. Primary keys cannot contain NULL values (Entity Integrity).
- Foreign Key: An attribute set in a referencing relation whose values must match the primary key of a referenced relation or be NULL (Referential Integrity).

2. Relational Algebra and SQL Queries

Relational algebra provides formal theoretical foundation for relational databases: Select (σ), Project (π), Union, Set Difference, Cartesian Product, and Natural Join (bowtie).

SQL translates these operations declaratively. Join types include:

- Inner Join: Returns rows when matching values exist in both tables.
- Left Outer Join: Returns all rows from left table and matched rows from right table (NULL filled if no match).
- Right Outer Join and Full Outer Join: Symmetric outer join variants.

CS103: Database Management Systems — Page 2: Normalization and Functional Dependencies

1. Functional Dependencies and Modification Anomalies

Poor database design induces update anomalies, insertion anomalies, and deletion anomalies. A Functional Dependency (FD) $X \rightarrow Y$ asserts that whenever two tuples agree on attributes X, they must agree on attributes Y.

Armstrong's Axioms: Reflexivity (if $Y \subseteq X$, then $X \rightarrow Y$), Augmentation (if $X \rightarrow Y$, then $XZ \rightarrow YZ$), and Transitivity (if $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$).

2. Normal Forms: 1NF, 2NF, 3NF, and BCNF

- First Normal Form (1NF): All attribute domains contain only atomic, indivisible values; repeating groups and multi-valued attributes are eliminated.
- Second Normal Form (2NF): Relation is in 1NF and every non-prime attribute is fully functionally dependent on every candidate key (no partial dependencies).
- Third Normal Form (3NF): Relation is in 2NF and no non-prime attribute is transitively dependent on a candidate key (for every $X \rightarrow A$, either X is a superkey or A is a prime attribute).
- Boyce-Codd Normal Form (BCNF): Stricter variant of 3NF. For every non-trivial functional dependency $X \rightarrow Y$, the determinant X must be a superkey.

CS103: Database Management Systems — Page 3: Transactions and ACID Guarantees

1. Transaction Definition and States

A transaction is a logical unit of database processing consisting of one or more read and write operations. A transaction must transition through discrete states: Active -> Partially Committed -> Committed, or if errors occur, Failed -> Aborted (followed by rollback).

2. The ACID Properties

- Atomicity: The 'all-or-nothing' principle. Either all transaction operations execute successfully to completion, or the database is rolled back to its pre-transaction state via Undo logs.
- Consistency: A transaction must transform the database from one valid state to another valid state, satisfying all declared schema constraints, unique keys, and foreign keys.
- Isolation: Concurrent transactions must execute without mutual interference. Intermediate uncommitted changes remain invisible to concurrent operations.
- Durability: Once a transaction commits, its modifications are permanently recorded in non-volatile storage and survive subsequent system crashes (enforced by Write-Ahead Logging WAL).

CS103: Database Management Systems — Page 4: Concurrency Control and Storage Indexing

1. Concurrency Control and Two-Phase Locking (2PL)

Uncontrolled concurrent access causes anomalies: Dirty Reads (reading uncommitted data), Non-repeatable Reads (data modified between reads), and Phantom Reads (rows inserted/deleted matching a query predicate).

Two-Phase Locking (2PL) protocol ensures conflict serializability:

- Growing Phase: Transaction may acquire locks (Shared S or Exclusive X) but cannot release any lock.
- Shrinking Phase: Transaction may release locks but cannot acquire new locks.

Strict 2PL: All exclusive locks are held until transaction commits or aborts, preventing cascading rollbacks.

2. Database Indexing: B+ Trees and Hash Indexes

- B+ Tree Index: Balanced multi-way search tree where all data records or record pointers reside exclusively in leaf nodes. Leaves are doubly linked to support fast range queries. Internal nodes store search keys and pointers, yielding high fan-out and shallow tree height (typically 3 to 4 levels for millions of records). Search, insert, and delete take $O(\log n)$ disk I/O.
- Hash Index: Uses a hash function to map keys to bucket blocks. Provides $O(1)$ point lookups, but does not support range scans or sorted ordering.